

# Dynamic Risk Posture Evaluation Using Automated Threat Intelligence

## Abstract:

One of the most urgent issues of contemporary enterprises is cybersecurity risk management, where manual, episodic assessments fail to consolidate real-time global threat intelligence with fine-grained technical telemetry. To address this, we describe DRPE (Dynamic Risk Posture Evaluation), a smart, fully containerized cybersecurity orchestration framework built using a high-performance FastAPI back-end, a React 18 enterprise dashboard, and a PostgreSQL relational database, coordinated by Docker Compose to be deployed in a single command.

Technically at the heart of the platform, DRPE uses a proprietary multi-variable Risk Engine which combines local technical telemetry of the OpenVAS/GVM infrastructure with global threat intelligence feeds of AbuseIPDB and AlienVault OTX. This generates a normalized, context-specific security posture score reflecting both intrinsic severity and real-world exploitability. Furthermore, the platform uses a Dynamic AI Engine with Google Gemini 1.5 Flash to support strategic decision-making, converting complex CVE identifiers and CVSS vectors to natural-language mitigation policies and executive-friendly impact assessments.

The system security is based on a stringent JWT authentication module alongside an immutable Activity Logging framework that provides auditability and proof of compliance. Client-side reporting (js PDF) generates executive compliance reports with negligible server load. Our experimental analysis indicates that the interactive 2D/3D Force-Directed Graph visualizations in the system significantly improve the speed and effectiveness of risk propagation analysis through network nodes as opposed to tabular lists of vulnerabilities.

## Group Members Detail:

|                  |                                  |              |
|------------------|----------------------------------|--------------|
| Zainab Khawar    | <a href="#">Linkedin profile</a> | 0332-1569733 |
| Adina Saif Ullah | <a href="#">Linkedin profile</a> | 0331-5113307 |

**Supervisor:** Dr. Faisal Bashir Hussain.  
Dean Faculty of Engineering Sciences.